

MOLECULAR PHYLOGENETICS & PHYLOGENOMICS — MOCK EXAM 2

A total of 27 closed questions, each worth one point, with no penalty for incorrect answers.

01. Phylogenomics is best defined as:

- A. The study of protein structure variations across distinct cellular organelles.
- B. The integration of genomic data and phylogenetic methods to infer evolutionary histories.
- C. A method that relies exclusively on mitochondrial DNA loops to classify organisms.
- D. The structural modeling of ancestral chromosomes inside extinct taxa.

02. The principle of Occam's Razor—asserting that the simplest explanation with the fewest assumptions is preferred—serves as the logical baseline for which tree reconstruction method?

- A. Maximum Likelihood
- B. Bayesian Inference
- C. Maximum Parsimony
- D. Neighbor-Joining

03. Choose the INCORRECT statement regarding the Neighbor-Joining (NJ) method:

- A. It utilizes a Q-matrix that adjusts raw distances based on net divergence parameters.
- B. It belongs to the category of character-based phylogenetic reconstruction algorithms.
- C. It does not assume a constant molecular clock, allowing lineages to evolve at distinct rates.
- D. It generates an unrooted tree topology by systematically matching pairs of nodes.

04. Which DNA substitution model allows nucleotide equilibrium frequencies to vary freely to fit compositional bias but assumes that all instantaneous transformation rates are completely equal?

- A. Jukes-Cantor (JC69)
- B. Kimura 2-Parameter (K80)
- C. Felsenstein 1981 (F81)
- D. General Time Reversible (GTR)

05. What type of molecular clock model operates under the explicit assumption that substitution rates are inherited by descendant branches and change gradually over time relative to parental lineages?

- A. Strict Molecular Clock
- B. Autocorrelated Relaxed Clock
- C. Uncorrelated Lognormal Clock
- D. Local Molecular Clock

06. In sequence alignment pipelines, the _____ algorithm computes optimal local matches to map isolated domains, while the _____ algorithm tracks complete end-to-end global alignments.

- 07. Which of the following represents a primary technical advantage of using RADseq or GBS approaches over full-genome sequencing?**
- A. It guarantees pristine assembly of multi-kilobase full-length coding gene regions.
 - B. It provides the absolute cheapest genome-scale marker screening, operating efficiently without reference templates.
 - C. It completely avoids errors associated with repetitive element misassemblies.
 - D. It explicitly maps the complete structural architecture of mitochondrial genomes.
- 08. The 'burn-in' phase is routinely executed and discarded at the start of a Bayesian phylogenetic MCMC chain to prevent:**
- A. The chain from getting trapped permanently inside the global maximum peak.
 - B. The over-parameterization of simple site models like Jukes-Cantor.
 - C. Unrepresentative random starting parameters from biasing the final stationary phase posterior estimations.
 - D. The duplication of in-paralogs inside the target orthogroups.
- 09. A taxonomic group that encompasses an ancestral sequence and some, but not all, of its descendant lineages is classified as:**
- A. Monophyletic.
 - B. Paraphyletic.
 - C. Polyphyletic.
 - D. Ultrametric.
- 10. Blomberg's K parameter is widely used to quantify phylogenetic signals in quantitative traits. A calculated value of $K > 1$ explicitly indicates that:**
- A. The trait has evolved completely randomly, resembling white noise independent of the tree.
 - B. The trait is significantly less conserved than expected under standard Brownian Motion models.
 - C. The trait is highly conserved, meaning closely related species look more similar than expected under Brownian Motion.
 - D. The trait is driven by massive directional positive selection across all terminal leaves.
- 11. Which heuristic tree rearrangement search strategy systematically clips out an entire subtree and regrafts it to every alternative branch to evaluate scores?**
- A. Nearest Neighbor Interchange (NNI)
 - B. Subtree Pruning and Regrafting (SPR)
 - C. Tree Bisection and Reconnection (TBR)
 - D. Reciprocal Best Rearrangement (RBR)
- 12. Homoplasy refers to sequence or phenotypic similarity among taxa that is driven explicitly by:**
- A. Direct inheritance from an immediate shared ancestral sequence containing synapomorphies.
 - B. Convergent evolution or independent evolutionary reversals back to an ancestral state.
 - C. Lineage-specific duplications leading to pristine single-copy orthogroups.
 - D. Stochastic variations occurring within low-temperature MCMC cold chains.

13. An 'orthogroup' is defined conceptually in comparative genomics as:

- A. A collection of genes that share structural identity but perform completely distinct biological functions.
- B. A group of homologous genes descending from a single ancestral gene in the last common ancestor of the studied species.
- C. An artificial assembly of terminal tips clustered exclusively by matching their GC content.
- D. A pair of genes within a single organism that originated from a tandem duplication event.

14. Over-partitioning a massive multi-locus supermatrix during a Maximum Likelihood tree search results in which of the following statistical issues?

- A. Forcing highly heterogeneous rates into a single biased parameter score.
- B. Overfitting the data by assigning excessive free parameters to short segments, collapsing statistical power.
- C. The complete elimination of transitions from the substitution Q-matrix.
- D. The transformation of phylograms into unrooted cladograms automatically.

15. What do site Concordance Factors (sCF) explicitly track along a specific internal branch of a reference tree topology?

- A. The proportion of single-locus gene trees that successfully recover that specific split.
- B. The percentage of individual informative alignment sites that support that specific branch split.
- C. The probability that the branch was generated during the MCMC burn-in phase.
- D. The absolute evolutionary time duration scaled by fossil calibrations.

16. Which of the following support metrics does NOT utilize a data resampling or deletion framework to compute branch values?

- A. Non-Parametric Bootstrap (BPs)
- B. Jackknife Support
- C. Approximate Likelihood Ratio Test (aLRT)
- D. Transfer Bootstrap Expectation (TBE)

17. In a classical substitution saturation plot, severe data saturation is visibly flagged by:

- A. A perfectly linear relationship between expected mutations and absolute sequence divergence.
- B. A clear curve plateau where transitions flatten out rapidly relative to transversions over time.
- C. The total collapse of the log-likelihood score down to zero.
- D. A symmetric bell-shaped distribution centered around an optimal phenotypic value.

18. The Robinson-Foulds (RF) distance parameter quantifies topological disagreement between two trees by counting:

- A. The total differences in absolute branch lengths across identical splits.
- B. The number of unique symmetric bipartitions (splits) present in one tree but missing from the other.
- C. The count of transitions required to transform one model Q-matrix into another.
- D. The frequency of incomplete lineage sorting events tracked within the nodes.

19. In character-based phylogenetic modeling, a/an _____ model implements explicit rules tracking biological mechanisms like selection, while a/an _____ model uses pre-estimated fixed substitution rates derived from large external databases.

20. Which of the following character features represents an autapomorphy of a lineage?

- A. An ancestral character state present across all members of a phylum.
- B. A unique derived trait confined exclusively to a single terminal taxon.
- C. A homoplasious structure shared by birds and bats due to adaptation.
- D. A point mutation that occurs inside an invariable site position.

21. The Approximately Unbiased (AU) test is highly recommended over the SH test because it uses a _____ bootstrap approach with variable block sizes, which effectively removes selection bias when tree topologies are evaluated _____ .

22. In a codon-based modeling framework, a branch-site model is explicitly engineered to detect episodes of positive selection targeting a restricted subset of _____ along specific designated _____ lineages.

23. In phylogenetic comparative methods, approaches like PIC and PGLS are mandatory because raw species data violate standard regression assumptions by exhibiting:

- A. Extreme heteroscedasticity driven by missing transcriptome values.
- B. Statistical non-independence due to shared evolutionary branch histories among the species.
- C. A perfectly lognormal distribution shape that prevents linear modeling.
- D. High levels of base composition saturation within terminal loops.

24. If the effective population size of an ancestral lineage is extremely small, the absolute occurrence of Incomplete Lineage Sorting (ILS) among descendant leaves will be severely _____ .

25. When using the General Time Reversible (GTR) nucleotide substitution model, the baseline framework incorporates a total of _____ independent base frequency parameters and _____ unique symmetrical substitution rate values.

26. A molecular clock model that enforces a uniform, static rate constancy within designated monophyletic groups but permits rates to vary freely between those groups is called a/an _____ molecular clock.

27. What type of distribution is most appropriate for modeling node calibration priors when geological barriers provide absolute hard maximum and soft minimum age boundaries?

- A. Uniform distribution
- B. Normal (Gaussian) distribution
- C. Lognormal distribution
- D. Symmetric binomial distribution

A SINGLE OPEN-ENDED QUESTION WORTH 5 POINTS, ANSWER IN MAX. THREE SENTENCES:

Describe the fundamental differences between systematic bias and stochastic bias in phylogenomics, and explicitly explain how increasing the total amount of sequence data (adding more genes) impacts each type of bias.
